



Full Length Article

Non-invasive real-time monitoring of cell concentration and viability using Doppler ultrasound

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ABSTRACT

Bioprocess optimization towards higher productivity and better quality control relies on real-time process monitoring tools to measure process and culture parameters. Cell concentration and viability are among the most important parameters to be monitored during bioreactor operations that are typically determined using optical methods on an extracted sample. In this paper, we have developed an online non-invasive sensor to measure cell concentration and viability based on Doppler ultrasound. An ultrasound transducer is mounted outside the bioreactor vessel and emits a high frequency tone burst (15 MHz) through the vessel wall. Acoustic backscatter from cells in the bioreactor depends on cell concentration and viability. The backscattered signal is collected through the same transducer and analyzed using multivariate data analysis (MVDA) to characterize and predict the cell culture properties. We have developed accurate MVDA models to predict the Chinese hamster ovary (CHO) cell concentration in a broad range from 0.1×10^6 cells/mL to 100×10^6 cells/mL, and cell viability from 3% to 99%. The non-invasive monitoring is ideal for single use bioreactor and the in-situ measurements removes the burden for offline sampling and dilution steps. This method can be similarly applied to other suspension cell culture modalities.

Introduction

Cell culture processing is the core of biomanufacturing, particularly in the market of monoclonal antibodies, which nearly doubled to \$163 billion between 2014 and 2019 [1]. To enable efficient and controlled automation in biomanufacturing, process analytical tools (PAT) are essential for the analysis of raw materials, in-process monitoring, and final product analysis. Despite the need and benefits of PAT, there is still limited commercial deployment in part due to a need for improved and online sensing technologies.

Cell concentration and viability are amongst the most important metrics understanding the health and productivity of a cell culture during biomanufacturing. Optical cell counters are commonly used for discrete measurements of cell concentration and viability [2]. A small volume of media (typically 10 μ L) is sampled and mixed with a viability dye (e.g., trypan blue), then transferred to a hemocytometer. Concentration is determined by extrapolating from the total number of cells counted in a fixed, known volume. Viability is then determined by the fraction

of those cells that have taken up the dye, an indication that the cell membrane is no longer fully intact [3]. Manually performing this procedure is extremely time-consuming and introduces human bias into the measurement. To address these drawbacks, multiple well-developed and commercially available devices work on this principle while partially or fully automating the mixing and counting steps such as Cedex HiRes (Roche CustomBiotech, Germany), Vi-CELL (Beckman-Coulter, USA), Eve (NanoEntek, Korea). Flow cytometers are also capable of extracting the cell viability and concentration by optically examining and counting the cells one by one in the flow [4]. Although these optical techniques are well established, the need to draw a sample for each measurement can be a significant drawback since each measurement may introduce the risk of contamination [5]. Further, this approach may be limiting or impractical for small-volume bioreactors where the total volume available for sampling is restricted. As an example, one measurement with the Cedex HiRes requires 300 μ L, or 2% of the 15 mL maximum volume of the Ambr 15 bioreactor (Sartorius Stedim, Germany). One solution is to run duplicate bioreactors at the same conditions and stagger sampling [6]. Moreover, these optical methods are not accurate in high concentrations above 10×10^6 cells/mL and require further dilution of the sample, which causes extra steps, and risks for contamination.

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To fully alleviate these concerns, in situ online sensing techniques are desired.

In situ microscopy has been demonstrated on large bioreactors, but this approach does not provide cell viability data or scale to small volume bioreactors [7]. Relatively new techniques have been developed based on Raman spectroscopy and capacitance sensors that allow such non-destructive, in situ measurements [8]. Models built on in situ measurements of Raman spectra with fed batch cultures have been shown to be predictive of viable and total cell densities in addition to other constituents of interest (glutamine, glutamate, glucose, lactate, ammonium) [9–12], and similar work is being conducted on perfusion systems [13]. However, Raman spectroscopy is not a plug-and-read optical sensor technology. Raman probes placed directly into the cell culture medium provide a molecular fingerprint relating to the vibrational spectroscopic information for all the molecular components within the system, including cells, media components and metabolites [14]. Therefore, the spectroscopic data often needs to be carefully modeled in a well-defined process. Raman techniques may also be impacted by fouling of the window over long run times [15].

In situ capacitance probe measurements have also been used to determine viable cell concentration in mammalian cell cultures [16]. These sensors are currently commercialized as online disposable sensors for example the Viamass by Sartorius Biotech GmbH. However, these sensors only measure the viable cell density and cannot measure the total cell concentration or the viability.

In this paper, we developed a non-invasive acoustic sensor based on ultrasonic pulsed Doppler (USPD) to monitor cell concentration and cell viability in real-time. Doppler ultrasound has been employed in several medical devices for ultrasonography and imaging [17] as well as measuring blood flow by bouncing high-frequency ultrasound waves off circulating red blood cells [18]. Takeda extended the method for characterization of flow and particles in two phase flow [19]. Africk and Colton used the Doppler ultrasound method to characterize pancreatic islets [20]. However, their method is limited to explicit analysis of the spectra characteristics such as peak frequency and amplitude and can only measure the concentration. We leveraged multivariate data analysis (MVDA) and developed prediction models with high accuracy to predict cell concentration and cell viability. Our sensor is comprised of an ultrasound probe that emits high frequency (15 MHz) ultrasound pulsed waves through the bioreactor vessel. The acoustic backscatter from insonified cells depends on their concentration and viability. The real-time method is non-invasive and can measure cell concentration up to 100×10^6 cells/mL without the burden of offline sampling or dilution.

Materials and methods

Instrumentation

Fig. 1 shows the schematic of the setup. A custom made focused ultrasonic transducer with a center frequency of 15 MHz and focal length of 6 mm is mounted to the exterior surface of a polycarbonate bioreactor (Ambr 15 Modular, Sartorius Stedim, Germany) with glycerin used as a coupling agent to improve impedance matching between the transducer and polycarbonate. The focal volume projects a few millimeters into the bioreactor media through a 1.2 mm thick side wall. Each bioreactor has an embedded impeller, which is used to stir the media and keep particles or cells suspended. To avoid acoustic reflections from the impeller, the transducer is mounted off-center. Temperature is controlled by a thermoelectric cooler module (CUI Devices, USA) attached to the bottom of the bioreactor and a Proportional integral derivative (PID) controller (ADN8831-EVALZ, Analog Devices Inc., USA).

The output of an arbitrary waveform generator (33522B, Keysight, USA) is tied to the transducer and the receive side of a pulser/receiver (UTPR-CC-35, TecScan, Canada). The waveform generator is configured to continuously send 20-cycle, 15 MHz tone bursts with an amplitude of 10 Vpp (peak to peak voltage) and burst period of 10 μ s. The transducer emits equivalent ultrasonic bursts into the bioreactor and during the quiescent period acts as a listener so that reflections from cells in the transducer path can be monitored. The pulser/receiver takes the combined signal and provides amplification, filtering, and clamping before passing along to a digital oscilloscope (MSO44, Tektronix, USA) where the signal is digitized. Both pulser/receiver and oscilloscope are synchronized to the waveform generator using the latter's SYNC output. After digitizing the signal, the oscilloscope performs spectral analysis and averaging. The final result is stored on a laptop computer for later processing and analysis. In addition to streamlining the collection procedures, the experiment is semi-automated by custom Python scripts running which can automatically adjust any configured parameters between measurements. The instrumentation is shown in Fig. 2 (a) and the test setup details in shown in Fig. 2 (b), which includes a motor to rotate the impeller of the Ambr 15 bioreactor, the temperature control unit, and the ultrasound probe.

Cell culture

CHO DG44 production clone was generated using Cellca CHO Expression Platform by Sartorius Sartorius Stedim Cellca GmbH. Cell cul-

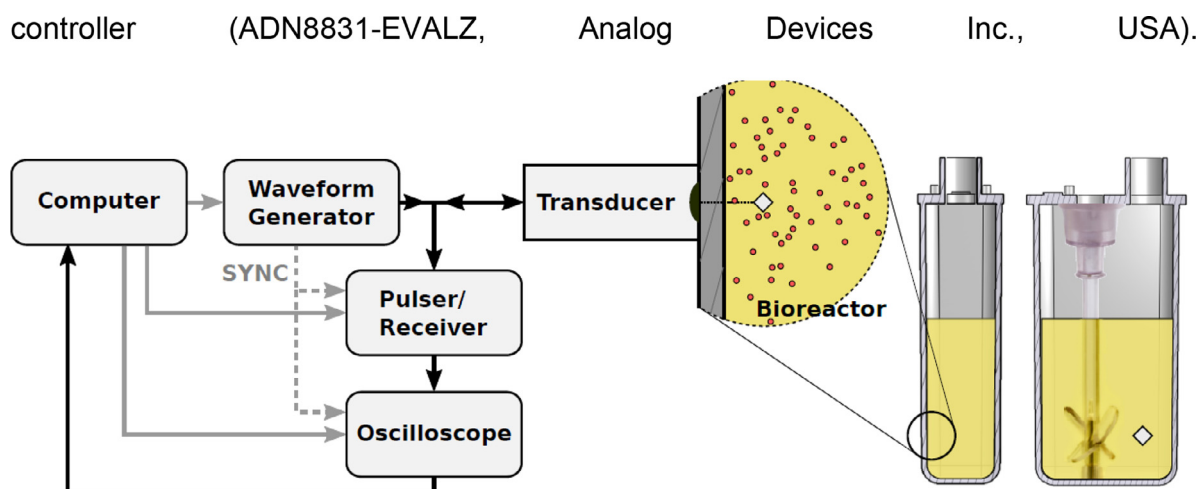


Fig. 1. Schematic of the Doppler ultrasound instrumentation. The ultrasound transducer mounted outside the bioreactor vessel and emits equivalent ultrasonic bursts into the bioreactor and during the quiescent period acts as a listener to collect the backscattered signal from cells in the bioreactor.

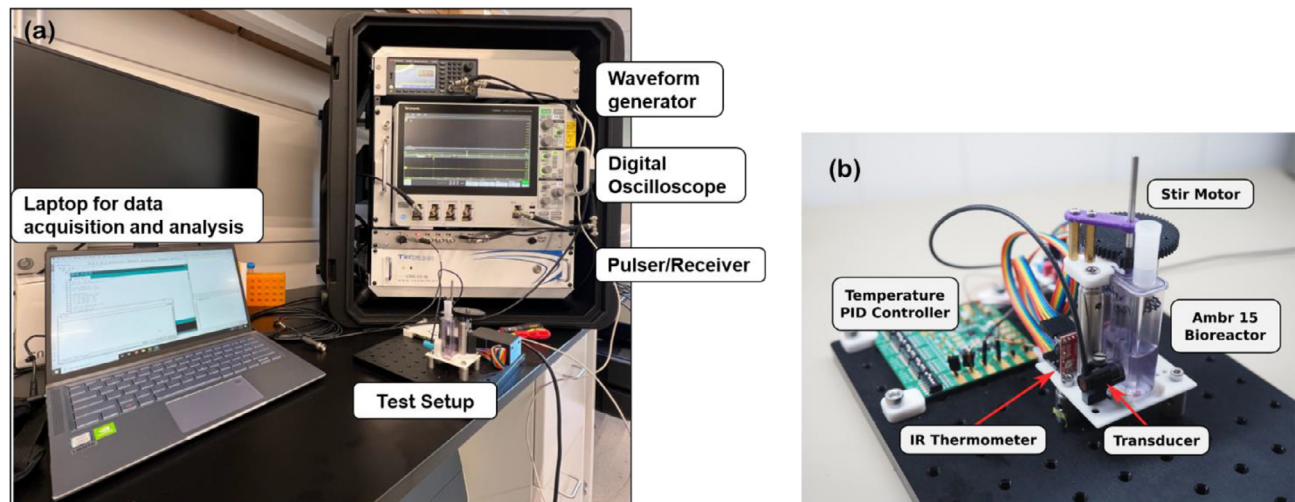


Fig. 2. Photo of the Doppler ultrasound setup. (a) shows the instrumentation including signal generation and data acquisition modules. The potential future commercial instrumentation can be significantly miniaturized. (b) Test setup mounted on the side of an Ambr 15 bioreactor. The ultrasound transducer touches the bioreactor vessel. An infrared thermometer measures the temperature and provides feedback to the PID controller to control the temperature in the vessel.

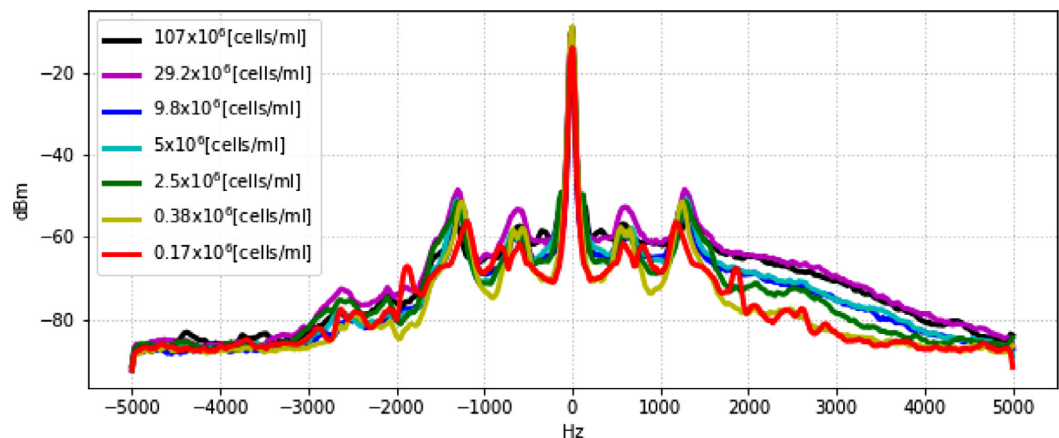


Fig. 3. Doppler ultrasound spectra for various samples with different cell concentration ranging from 0.1 to 100×10^6 cells/mL. Cell viability of all samples is $>98\%$. X axis is the spectral frequency subtracted from the ultrasound frequency (15 MHz). Y axis is the amplitude of the spectrum.

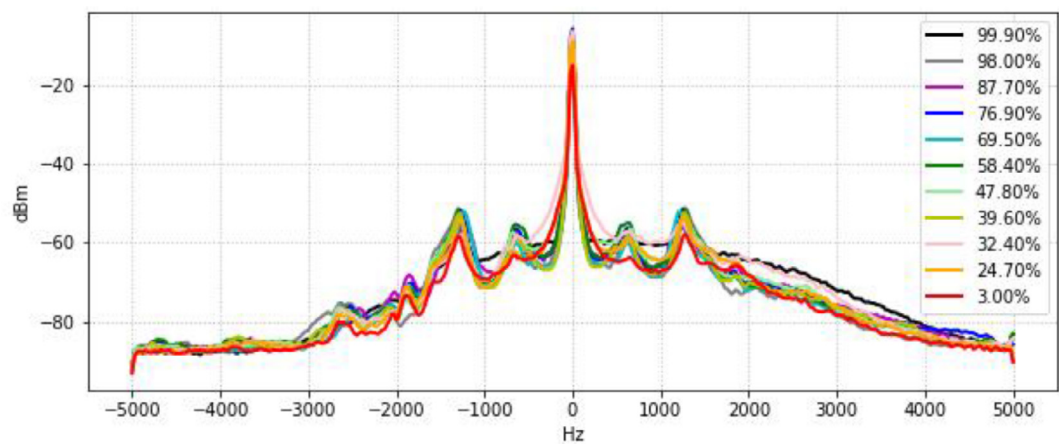


Fig. 4. Doppler ultrasound spectra of samples with different cell viability ranging from 3% to 99%. The cell concentration is kept constant at 5×10^6 cells/mL. X axis is the spectral frequency subtracted from the ultrasound frequency (15 MHz). Y axis is the amplitude of the spectrum.

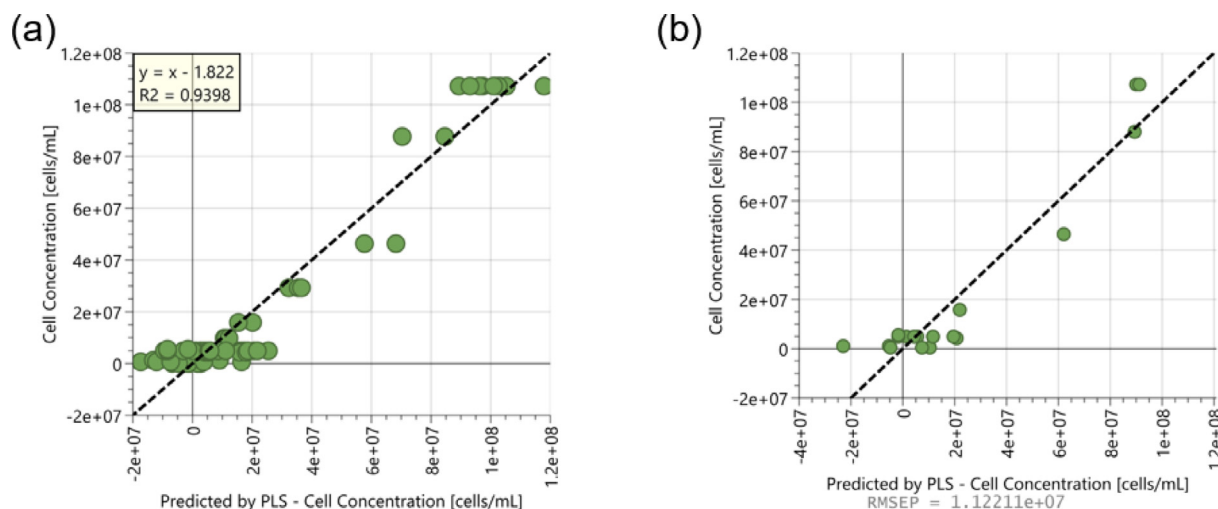


Fig. 5. Single PLS model is not accurate in prediction of the very high and low cell concentration. (a) Prediction of 72 observations used in model development. (b) 18 observations used for model validation.

ture media was composed of Cellca SMD part # 72739C (SAFC, Merck KGaA, Darmstadt, Germany) and 6 mM L-glutamine (Corning, NY, USA). Cells were cultivated by standard method. Briefly, 10×10^6 cells were dispersed in 50 mL media to achieve inoculation concentration of 0.2×10^6 cells/mL in a 250 mL shake flask. Cells were incubated at 37.0°C with 7.5% CO_2 supplement and an orbital shaking at 103 rpm and passaged every 3 days.

Cell concentration study with high viability

Cells were maintained at high viability (>98%) by standard cultivation method. A dilution method was used to prepare a range of cell concentrations from 1.68×10^5 to 1.07×10^8 cells/mL. Cells were pelleted at 190 g for 3 min and redispersed in 9 mL fresh media in a 15 mL tube to obtain the highest cell concentration. 300 mL cell solution was used to determine the cell concentration and viability by Cedex HiRes Analyzer (Roche Diagnostics GMBH, Germany). Then 8 mL of cell solution was transferred to an Ambr 15 vessel for the measurements. After the measurements, cells were transferred to a 15 mL tube and diluted with media to obtain 9 mL cell solution at lower concentrations and the Cedex measurements were repeated. Each concentration of cells was measured 3 times at 1050 rpm stir rate and room temperature ($24 \pm 1.5^\circ\text{C}$).

Cell viability study with fixed cell concentration

Cells with lower viability were obtained by reducing gas supplement. On the passage day, when the cells reached high concentration, the fil-

ter cap of the shake flask was wrapped with parafilm to block air flow. Cell viability decreased subsequently and finally reached nearly 0%. In this study, cell concentration was kept constant at $5 \times 10^6 \pm 0.27 \times 10^6$ cells/mL for all viabilities. To prepare samples with a range of cell viabilities from 3% to 99.6 %, cells at the highest viability were pelleted and redispersed in 9 mL fresh media in a 15 mL tube. Cedex measurements were performed using 300 mL cell solution and then 8 mL cell solution was transferred to an Ambr 15 vessel for the Doppler ultrasound sensor measurements. After the measurements, cells were transferred to a 15 mL tube and mixed with cells at lowest viability at calculated ratios to obtain 9 mL cell solution at lower viability. Cedex measurements and the Doppler ultrasound measurements were repeated for each condition as described previously.

Supplementary Table 1 and Table 2 contains the details of samples used with the Ambr 15 setup for Doppler ultrasound measurements.

Combined cell concentration and viability study

In this study, we varied both cell concentration and cell viability. Samples with cell concentration within the range 0.2×10^6 to 9.6×10^6 cells/mL and cell viability of 2% to 98% were tested with the Doppler ultrasound setup.

To prepare samples, two cell culture flasks were maintained at high viability (>98%) by standard cultivation method, and two separate cell culture flasks were maintained at low viability (<10%), following the same procedure described in the cell viability study. Both cultures were pelleted and redispersed to obtain the highest cell concentration, and then 10 mL cell solution was used to determine concentration and vi-

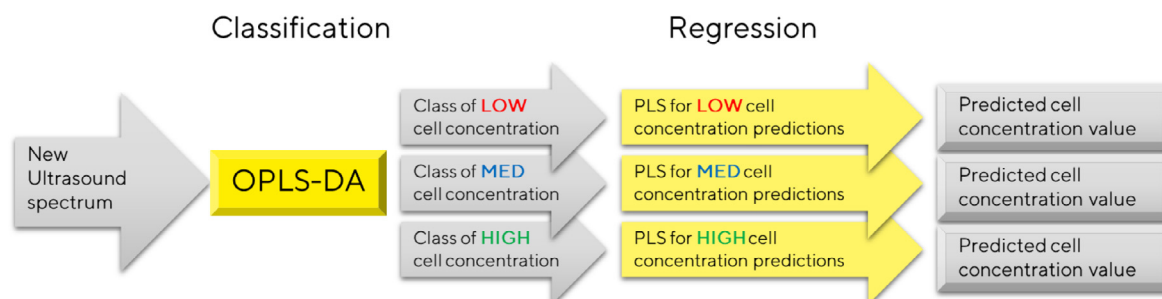


Fig. 6. Schematic of two step classification/regression model for high accuracy prediction of cell concentration in the broad range of $0.1\text{--}100 \times 10^6$ cells/mL

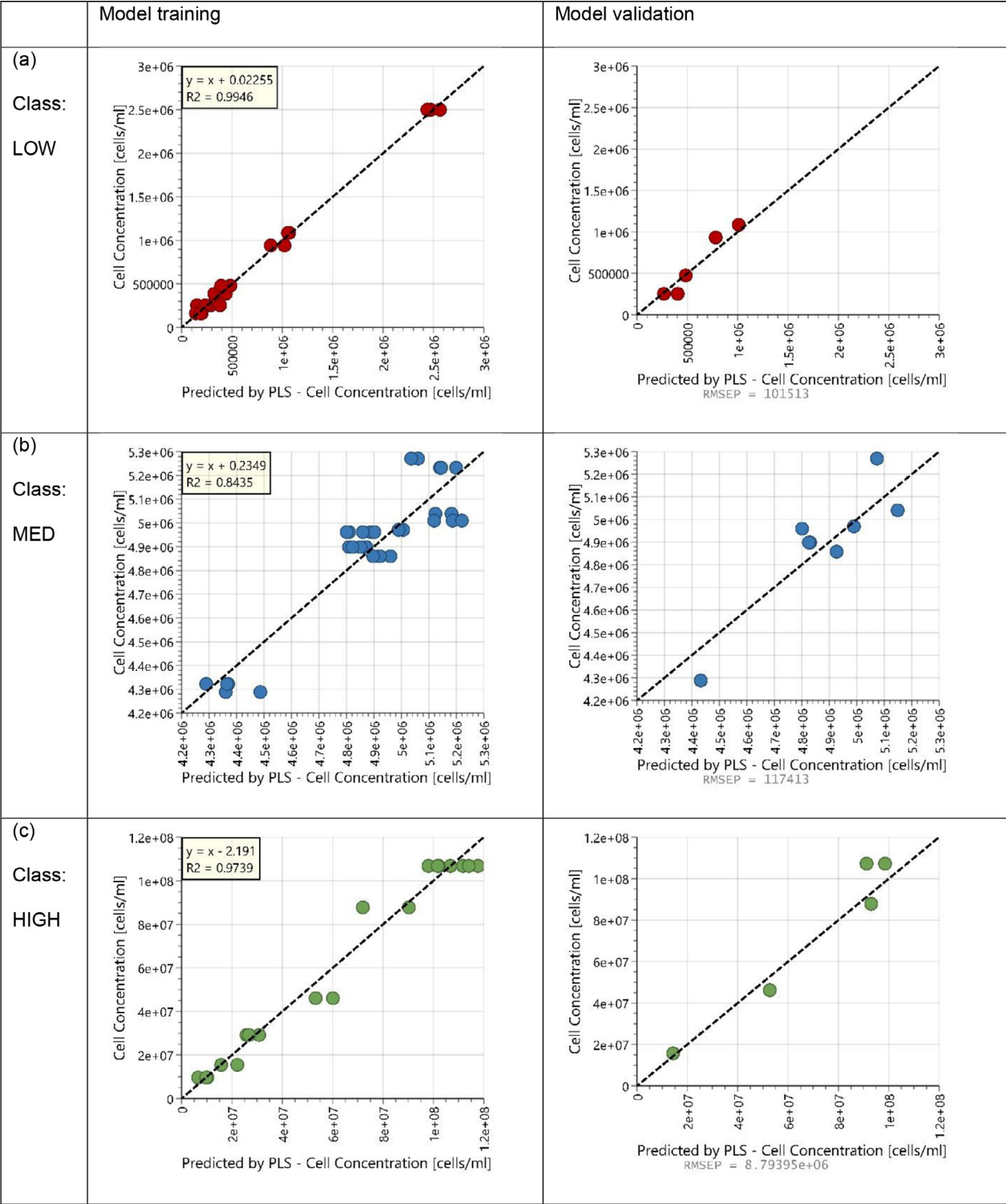


Fig. 7. PLS model prediction values against the cell concentrations using PLS models fitted for the three sub-classes. The data shown in the graph in the second column are not seen by the trained model and are used for validation study. (a) Class LOW, is for the low cell concentration class $<3.0 \times 10^6$ cells/mL, (b) Class MEDIUM is for the medium cell concentration class between 3.0 and 9.0×10^6 cells/mL, and (c) Class HIGH is for the high cell concentration class $>9.0 \times 10^6$ cells/mL.

ability of both by Eve Automated Cell Counter (NanoEnTek, South Korea). 8 mL of solution was transferred to an Ambr 15 vessel for measurements, with the ratio of high to low viability cultures varied to achieve different target viabilities of the mixture. Each mixture was tested at a range of concentrations by serial dilution with fresh media,

with concentration and viability remeasured at each dilution step using the Eve Automated Cell Counter. The Doppler ultrasound measurement on these samples were measured 3 times at 1050 rpm of stir rate and at room temperature. The spectra is collected and analyzed with SIMCA.

MVDA data analysis methods

We used spectroscopy module of SIMCA V.17 software (Sartorius Stem Data Analytics AB, Sweden) and provided the Doppler ultrasound spectra as the input. The PLS (partial least square) regression and OPLS-DA (orthogonal projections to latent structures discriminant analysis) modules built in SIMCA were used for modeling. For PLS regression prediction model, cell concentration and viability were considered as the output. For the PLS and OPLS-DA models, mean-centering was performed over the whole range of captured spectral wavelength. NIPALS [23] algorithm of SIMCA fitted the models. Around 80% of the observations were used for training the model and the rest were kept for validation.

Results & discussion

The Doppler ultrasound waveform of each sample at different cell concentration and viability are captured via the oscilloscope and then Fourier transformed using a Hanning filter, and the process is repeated 256 times to build up an average spectrum. Frequencies outside of a 10 kHz band centered at the interrogation frequency (15 MHz) are discarded. The spectral graph of samples with different cell concentrations is shown in Fig. 3. Cell concentration of samples ranges from 0.1×10^6 to 100×10^6 cells/mL and the viability is $>98\%$ for all samples. Higher cell concentrations result in more backscattered power, as indicated by the larger area under the curve in the spectra. The base shape of the spectrum and the number of peaks depends on the transducer and the vessel shape and configuration.

Fig. 4 shows the spectral graph of samples with cell viability ranging from 3% to 99% and cell concentration kept constant at 5×10^6 cells/mL. The spectral shape of the samples change with viability; however, there is no obvious correlation between the cell viability and the spectra's main characteristics such as the frequency or amplitude of a particular peak. Thus, we leveraged multivariate data analysis (MVDA) to create prediction models for these variables. MVDA is a statistical technique suitable and powerful to analyze the data generated from multiple variables, specially, when the input data are colinear and coupled. We have used the whole spectrum as the input for the MVDA model using the spectroscopy module of SIMCA.

We used partial least square (PLS) regression to fit a model to correlate the cell concentration to the measured Doppler ultrasound spectra. PLS is one of the standard methods used in the field of multivariate calibration to fit a linear equation for the regression problem [21]. We used

80% of the dataset to train and create the model and kept the rest, not initially seen by the model, to validate the accuracy of the model. The observations used to train the PLS model is shown in Fig. 5(a) and the observations not seen by the prediction model used for validation are shown in Fig. 5(b). The single PLS model is not very accurate for prediction of low and high cell concentrations as the observations get parted from the model.

To further improve the prediction accuracy, we developed a two-step model including first classification into 3 categories of low, medium and high cell concentration and then PLS regression in each category to accurately predict the broad range of cell concentration, as shown schematically in Fig. 6. Using the orthogonal projections to latent structures Discriminant Analysis (OPLS-DA) built-in SIMCA, we divided the overall cell concentration into three classes of LOW ($<3.0 \times 10^6$ cells/mL), MED (3.0×10^6 cells/mL $< 9.0 \times 10^6$ cells/mL) and HIGH ($>9.0 \times 10^6$ cells/mL). OPLS-DA is a powerful statistical tool that provides insight into separation between spectral measurements.[22] Validation of a fitted OPLS-DA model against samples unseen by the model demonstrated high capability of the model for accurate classification of new samples (sensitivity and specificity of 100%), as shown in Supplementary Fig. 1. Then, a PLS regression model was fitted to each one of the classes to predict CHO concentration, as shown at Fig. 7.

We compared the performance of the two-step classification/regression model with the one overall PLS model fit to the whole dataset. Root mean square error of predictions (RMSEP) for the overall PLS model outputs (PLS_Pred_1) and for the outputs of the three PLS sub-models of classes (PLS_Pred_2) are tabulated in Supplementary Table 3. For validation samples of the LOW concentration class, RMSEP was reduced from 1.26×10^7 to 1.02×10^5 (99.19% reduction) in favor of the two-step modeling approach. In similar manner, RMSEP values for MED and HIGH concentration classes were deducted 98.67% and 32.18% respectively.

For cell viability prediction, we prepared samples with viability of 3% to 99% by mixing the dead and highly viable cells. The Doppler ultrasound spectra from the instrument were used for training the model. We used 80% of the dataset for training and kept 20% for validation study. Fig. 8(a) shows prediction of 29 observations used for model training, and Fig. 8(b) shows the 7 observations not seen by prediction model and used for model validation. We found that a single PLS regression model can nicely predict the cell viability.

In the studies above, we demonstrated accurate prediction of the cell concentration for a highly viable cell culture and accurate prediction of cell viability while the cell concentration was constant around

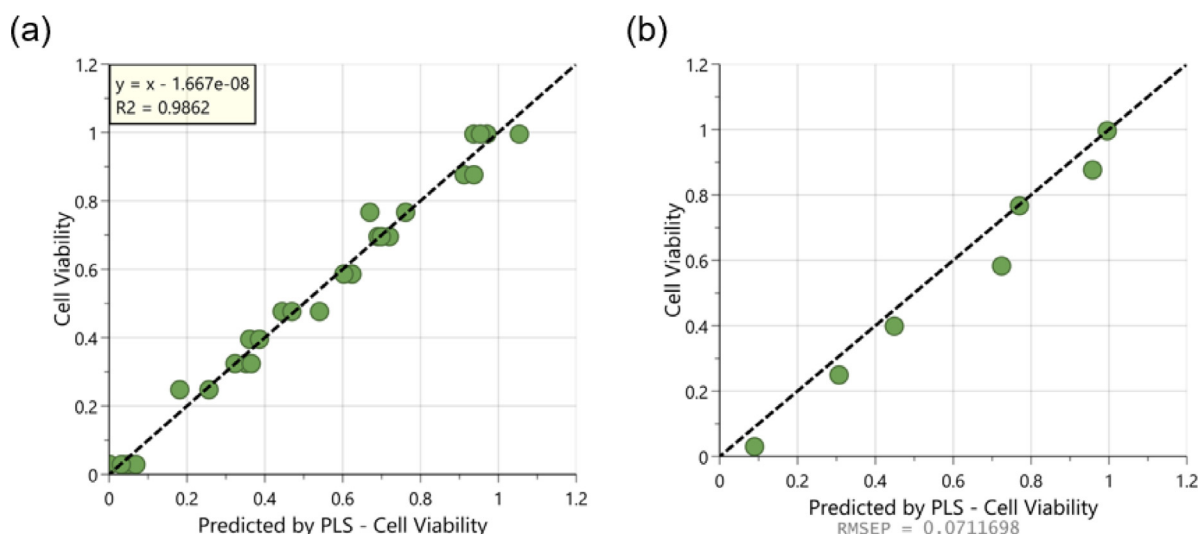


Fig. 8. Cell viability of samples accurately predicted using the PLS model. (a) Prediction PLS model and 29 observations between 3% and 99% viability. (b) Validation model with 7 observations not seen by the prediction model.

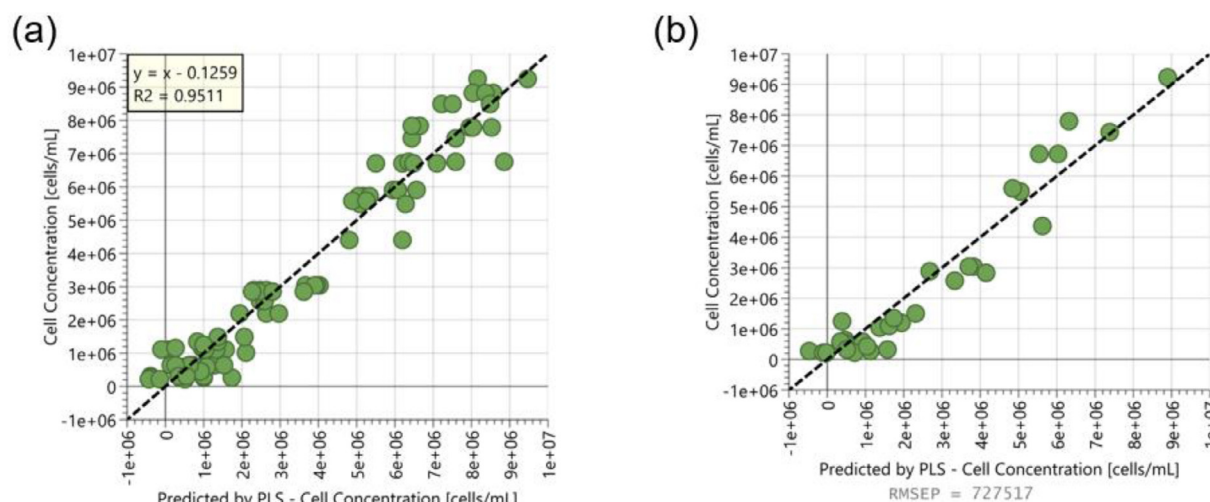


Fig. 9. Cell concentration of samples predicted using the OPLS model. The samples had cell concentrations ranging from 0.2×10^6 cells/mL to 10×10^6 cells/mL and cell viability ranging from 2 % to 98%. (a) Prediction OPLS model. (b) Validation model using the observations not seen by the prediction model.

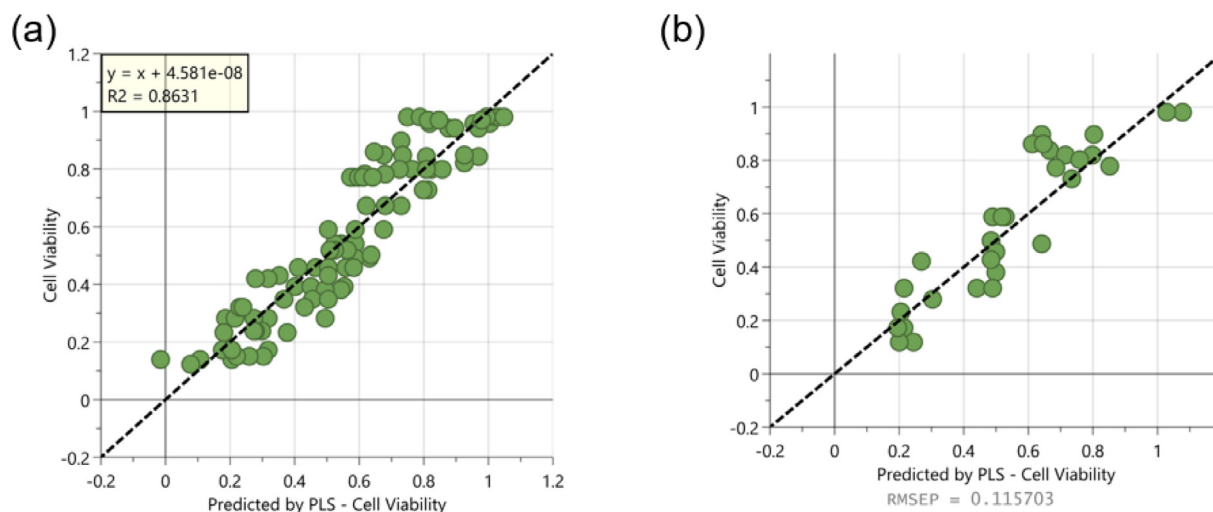


Fig. 10. Cell viability of samples predicted using the OPLS model. The samples had cell concentrations ranging from 0.2×10^6 cells/mL to 10×10^6 cells/mL and cell viability ranging from 2 % to 98% (a) Prediction OPLS model. (b) Validation model using the observations not seen by the prediction model.

5×10^6 cells/mL. As the next step, we made samples with varying cell concentration and cell viability within relevant bioprocessing ranges. For that, we mixed highly viable cells (viability of 98%) and mostly dead cells (viability ~2%) at desired volume ratios to make samples with varying concentration and viability, as listed in Supplementary Table 3. The cell concentration and viability were then measured via Eve optical cell counter (NanoEntek, Korea), as a reference for prediction. The Doppler ultrasound spectra of the samples were then captured and used to train MVDA models. Two OPLS models were created for cell concentration and cell viability. Similar to previous studies, 80% of the observations were used for training and 20% of the observations that were not seen by the training models were kept for model validation. Figs. 9 and 10 show a nice fit for cell concentration and cell viability, respectively.

In conclusion, we have developed an in situ and non-destructive ultrasound-based sensor for real-time monitoring of cell concentration and cell viability. Using our classification-regression MVDA technique, we could predict a wide range of cell concentration from 0.1 to 100×10^6 cells/mL, which is a great improvement compared with optical methods that are not accurate at high cell density cell culture $>1 \times 10^7$ cells/mL. This sensor can be similarly expanded for other cell culture modalities including HEK, T cells and stem cells and will poten-

tially improve the bioprocess development providing real-time information on cell health and growth. The non-invasive monitoring of the cell culture with the transducer mounted outside the bioreactor makes it easy to implement for single use assemblies including vessels and bags.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.slast.2022.09.003](https://doi.org/10.1016/j.slast.2022.09.003).

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